

Problem set 3 KNN

Exercise 1

Consider the following labelled training points in R^2 :

Point	(x, y)	Class
A	$(1, 1)$	Red
B	$(2, 1)$	Red
C	$(4, 3)$	Blue
D	$(5, 4)$	Blue

Using the k -**Nearest Neighbours** classifier with $k = 3$ and **Euclidean distance**, determine the predicted class of the test point

$$Q = (3, 2).$$

Show your distance calculations and the majority vote among the k nearest neighbours.